A Falsifiable Closure Hypothesis Beneath Binary Quantum Readout

Two-face pairing, eight configurations, no-go boundary, and staged experimental protocols

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Abstract. We formulate a candidate closed-structure layer beneath ordinary binary quantum readout. The proposal separates three levels: a determinate binary readout, the usual quantum-state description, and a hypothetical directional closure structure. A newly transcribed hand-specified table contains sixteen raw faces organized into eight pairs: at each of four slots the two faces have the same direction label and complementary binary value. This establishes a reproducible 16-to-8 classification rule, but an inversion audit shows that a single label such as 0_A leaves four candidate closure pairs, and even a slot-resolved 0_A leaves two. Thus the hypothesized label does not yet give a unique reconstruction; a second independent readout or a frozen dynamical rule is required. We prove a basic operational boundary: if distinct preparation histories yield the same density operator and all subsequent operations obey standard completely positive trace-preserving maps and POVM measurements, then no path-dependent signal can occur. We give three staged protocols with conventional adversaries and explicit failure conditions. The package is a conservative invitation to test a sharply delimited hypothesis, not evidence that sub-quantum structure has been observed.

Keywords: quantum foundations; operational equivalence; preparation contextuality; process memory; hidden effective dimension; quantum instruments; falsifiable hypothesis; closure structure

1. Motivation and scope

A binary measurement returns 0 or 1. Standard quantum theory describes the pre-measurement system by a density operator rho, including superpositions of computational-basis states. This work studies a narrower speculative question: could the ordinary quantum description itself be a coarse-grained description of a directional closed structure? A visual intuition such as 0 -> (0,1) and 1 -> (1,0) is not used here as a literal decomposition into two ordinary qubits. It is used only as a candidate relational notation for opposite directions within a proposed underlying structure.

The proposal is deliberately separated from claims about consciousness, cosmology, particles below experimentally established scales, or a completed replacement for quantum mechanics. The present document offers neither. Its contribution is a falsifiable architecture: define the hypothesized variables, state exactly what standard quantum mechanics forbids under operational equivalence, and specify what data would count only as memory or leakage rather than as evidence for a new layer.

2. Three-layer closure hypothesis

<table>
<tr><th>Layer</th><th>Working description</th><th>Status</th></tr>
<tr><td>I</td><td>Determinate binary readout x in {0,1}.</td><td>Established measurement language.</td></tr>
<tr><td>II</td><td>Ordinary quantum state rho; for a pure qubit, |psi> = alpha|0> + beta|1>.</td><td>Standard quantum theory.</td></tr>
<tr><td>III</td><td>Candidate closure configuration lambda in Lambda = {lambda_0,...,lambda_7}, provisionally organized as four directions times two faces.</td><td>Hypothesis only; no independent physical evidence.</td></tr>
</table>

A useful provisional representation is Lambda congruent to Z_4 x Z_2. The Z_4 factor labels a cycle of four directions, while Z_2 labels a two-face reversal. This representation classifies eight candidate configurations. It does not, by itself, establish that nature contains eight new physical states, nor does a finite-cycle identity such as R^8 = I distinguish the hypothesis from conventional finite-state or higher-dimensional quantum constructions.

2.1 Handwritten two-face table and inversion audit

The supplied handwritten source fixes eight front faces. Each front face has four ordered slots (top-left, top-right, bottom-left, bottom-right), and each slot contains a binary value with a direction label A, B, C, or D. The paired reverse face is defined by preserving the slot and direction label while complementing the binary value. This gives 16 raw faces grouped into 8 closure pairs. The complete machine-readable transcription and audit script are included in this package.

Closure pair	Front face: TL, TR, BL, BR	Reverse face rule
lambda_0	0_A, 1_B, 1_C, 0_D	same labels; 0 <-> 1 in every slot
lambda_1	0_B, 1_A, 1_D, 0_C	same labels; 0 <-> 1 in every slot
lambda_2	0_C, 1_D, 1_A, 0_B	same labels; 0 <-> 1 in every slot
lambda_3	0_D, 1_C, 1_B, 0_A	same labels; 0 <-> 1 in every slot
lambda_4	0_C, 1_A, 1_D, 0_B	same labels; 0 <-> 1 in every slot
lambda_5	0_B, 1_D, 1_A, 0_C	same labels; 0 <-> 1 in every slot
lambda_6	0_D, 1_B, 1_C, 0_A	same labels; 0 <-> 1 in every slot
lambda_7	0_A, 1_C, 1_B, 0_D	same labels; 0 <-> 1 in every slot

Audit result. Across the eight front faces, each intrinsic label (for example, 0_A) occurs in four candidates. Adding its top-left location reduces 0_A to lambda_0 or lambda_7, but does not select one uniquely. Therefore the table supports the two-face/eight-pair classification, while it falsifies the stronger present claim that a single observed label can by itself reconstruct the entire configuration. Stage III must include a second independent readout or a fully specified transition rule.

3. Operational no-go boundary

Proposition 1 (same-state indistinguishability under standard closure). Let P_k be preparation procedures such that rho_k = rho_star for every k. Let E be a k-independent CPTP map and {M_x} a k-independent POVM. Then p(x|P_k) = Tr[M_x E(rho_star)] is independent of k.

Proof. By premise, every preparation produces the same state rho_star in the ordinary quantum description. Applying a path-independent channel and then a path-independent measurement produces the same Born-rule expression for every k. Hence the probabilities cannot depend on the preparation label. QED.

This is the first death gate. If a same-state/different-history experiment yields a difference, the immediate possibilities are: incomplete state equivalence, an environment carrying memory, population leakage outside the modeled system, drift or state-preparation-and-measurement error, a history-dependent control path, or a departure from standard operational quantum theory. The last possibility is not licensed until the preceding standard explanations have been tested as serious competitors.

4. The value-plus-phase claim

The sharpened hypothesis is not merely that there are eight configurations. It is that a binary readout could have the form (x,phi), where x is the ordinary value and phi is an intrinsic closure position or phase. The word intrinsic matters: phi must not be an instrument angle, a pulse label, a time bin, a software branch, or an analysis tag.

There are two logically distinct cases. If phi is directly read out, the device has an eight-outcome result alphabet, and the scientific question becomes whether this refinement is reducible to a standard higher-dimensional state, a leakage process, or a memory-bearing environment. If phi is not directly read out, then a reported 0 cannot be silently promoted to 0_A. The model must instead predict a second, fixed probe whose conditional statistics reveal phi without using the preparation record.

A valid closure inversion rule must therefore be a table of pre-registered conditional probabilities p(y,e | x,d; W_2), where W_2 is a fixed second probe, d is a direction outcome if experimentally available, and e is a second result. Qualitative wording such as 'the remaining positions persist' is not enough for a physical test.

5. Staged experiments

Stage	Question	What a positive result means	What it cannot establish alone
I	Can identical apparent qubit states prepared through different histories yield reproducible future differences?	Process memory, hidden effective dimension, or a device-level anomaly worth modeling.	A sub-quantum closure layer.
II	Does a pre-specified four-direction/two-face rule yield an eight-cycle conditional pattern beyond standard competitors?	An eight-configuration model has predictive value under the registered resource constraints.	That the eight configurations are ontologically fundamental.
III	Does a putative intrinsic label (x,phi) make correct blind predictions of a second probe while ordinary rho is held fixed?	A candidate operational extension requiring independent replication.	A discovery of smaller material constituents.

5.1 Stage I: microscopic closure versus ordinary memory

Prepare two or more operationally matched final qubit states through distinct control histories. Audit the full pulse sequence, timestamps, randomized ordering, calibration records, raw shots, and tomography. Apply a fixed recovery or interrogation sequence not used in fitting. Compare against a no-memory qubit channel, finite-environment models, qutrit/ququart leakage, process-tensor models, SPAM and drift, and classical history-dependent control. This experiment is useful immediately because it can diagnose memory; it is not a test of the 0 -> 01 notation unless the closure model supplies a prediction that the competitors cannot reproduce.

5.2 Stage II: discrete closure orientation test

The hand-specified 8 x 4 table is now available in Section 2.1. Define eight physical preparation paths and a fixed probe. Pre-register the entire eight-component signal vector S_k = 2p(1|P_k,W)-1, including sign, amplitude, phase convention, and allowed shared parameters. A valid result must survive blind confirmation and outperform standard process, leakage, and drift models with comparable complexity. This stage remains a design template because the physical map from the table to a numerical signal vector has not yet been derived.

5.3 Stage III: value-plus-phase and inversion test

Only after Stage II yields a registered, reproducible pattern should an intrinsic-label test be attempted. The first measurement must produce either a directly observed refined outcome (x,phi) or a reproducible proxy inferred without using the preparation record. The hand-table audit shows that one label is insufficient for unique inversion, so the model must predict a second-probe conditional table that distinguishes the remaining candidates. The main endpoint is a blind likelihood or Bayes-evidence advantage for the frozen table over the adversaries. A result that appears only after selecting a favorable second probe, direction, time slice, or phase convention is a failure of the protocol.

6. Required controls and failure criteria

Threat	Mandatory control	Failure interpretation
State mismatch	Independent tomography and conservative distance bound before the final probe.	Ordinary different-state effect.
Environment memory	Process-tensor or finite-environment competitor with held-out prediction.	Memory, not a closure layer.
Leakage	Leakage-sensitive readout or explicit qutrit/ququart competitor.	Higher effective dimension.
Drift/SPAM	Interleaving, randomized path order, calibration logs, and null paths.	Control artifact.
Post-selection	Freeze endpoints, exclusions, and analysis before blind data.	Exploratory observation only.

7. Relation to existing frameworks

Operationally equivalent preparations with different proposed underlying descriptions sit near the literature on preparation contextuality. Spekkens developed a general operational framework for contextuality across preparations, transformations, and unsharp measurements. Process-tensor tomography provides a standard quantum framework for testing multi-time memory. These are not peripheral citations: they are the first serious adversaries for the present program. The closure hypothesis must either make a more restrictive, successful prediction or remain an interpretive relabeling of standard process structure.

8. Current status and conclusion

The present work has established a disciplined research program, not a physical discovery. The hand-specified sixteen-face table now supplies a reproducible 16-to-8 pair classification and exposes an important limit: a single value-plus-direction label does not uniquely invert the closure state. The operational same-state no-go boundary is exact under standard quantum closure. Stage I is already scientifically meaningful as a memory/hidden-dimension diagnostic. Stages II and III remain contingent on a missing derivation: a coarse-graining map to rho and a fixed numerical signal or second-probe conditional-probability table not freely adjustable per device. The appropriate next action is to derive those objects, followed by an openly preregistered experimental card. If that completion fails, the correct conclusion is not that the idea was ignored, but that it did not yet yield an operational physical theory.

References

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